

XAVIER FRANKLINE ODHIAMBO

xavierfrank4@gmail.com | xodhiambo.com | github.com/theepsalmist | +254786409255 | Remote

Summary

- Senior software engineer with 7 years of Python experience building production web applications across SaaS, civic-tech, and distributed data platforms. Strong on API engineering (Django REST Framework, FastAPI), database design and optimisation (PostgreSQL, ClickHouse, Elasticsearch), cloud infrastructure (AWS Certified Solutions Architect), and operating distributed systems through CI/CD, observability, and incident response. Recent work includes production AI engineering: RAG pipelines, LLM orchestration infrastructure, and self-hosted model deployment. Open-source contributor to Django and CKAN.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash

Backend & APIs: Django, Django REST Framework, FastAPI, Flask, Celery, RESTful API design

Databases: PostgreSQL, MongoDB, ClickHouse, Elasticsearch, Redis

Cloud & Infrastructure: AWS (EC2, RDS, ECS, Lambda, S3, Route 53, SNS), Docker, Kubernetes, Terraform, Ansible

CI/CD & Tooling: GitHub Actions, GitLab CI/CD, Bitbucket Pipelines, Git

Observability: OpenTelemetry, Grafana, Loki, Tempo, Prometheus, Sentry, CloudWatch

AI Engineering & Tools: LlamaIndex, pgvector, OpenRouter, Ollama, OpenWebUI, Claude Code, Cursor

Certifications

AWS Certified Solutions Architect – Amazon Web Services, January 2025

AWS Certified Cloud Practitioner – Amazon Web Services, January 2022

Experience

Senior Software Engineer, Search Atlas (Remote)

August 2025 – June 2026

Backend engineer on an SEO analytics platform integrating Google Search Console, Google Analytics, Google Ads, and Meta Ads for advertising agencies and in-house marketing teams.

- Designed and built REST API services in Django REST Framework with strong input validation, structured error handling, authentication, and rate limiting, backing both internal services and a customer-facing analytics product.
- Wrote and shipped PostgreSQL and ClickHouse migrations, optimised slow queries through indexing and schema refactors, and worked on partitioning strategies for high-volume analytics tables.
- Owned the email delivery and monitoring component of the platform's reporting microservice, adding AWS SES event tracking for opens, clicks, and bounces, giving agencies visibility into delivery health and resolving a recurring source of client complaints about undelivered emails.
- Instrumented services with OpenTelemetry across Django and Celery; resolved production incidents by correlating Sentry exceptions, Loki logs, and Grafana metrics.
- Improved and extended the platform's LLM orchestration layer, adding multi-provider routing, automatic failover, and cost-aware model selection for internal services handling automated analysis and content generation.

Senior Software Engineer, Code for Africa (Remote)

August 2022 – August 2025

Backend and infrastructure engineer building systems for civic-tech platforms used by researchers, journalists, and policy teams across Africa.

- Built and maintained Django REST Framework, FastAPI, and Flask APIs across multiple platforms, including distributed news indexing, IoT sensor ingestion (Sensors.Africa), and census/demographic data platforms (Wazimap-NG).
- Led the architecture and deployment of multi-node Elasticsearch clusters using Ansible automation, including a Rally-based acceptance benchmark harness that validated indexing throughput, query latency under load, and mixed read/write performance before newly provisioned clusters accepted production traffic.
- Implemented end-to-end observability with Grafana, Prometheus, Loki, and Alertmanager, investigating and resolving production incidents by correlating Sentry exceptions, logs, and metrics.
- Drove adoption of CI/CD pipelines across engineering teams using GitHub Actions and GitLab CI/CD, with automated testing, containerised image publishing, and technical documentation as standard practice.

- Led development of an internal ad-quality and domain-trust classifier, a gated multi-stage RAG pipeline combining automated signal checks with an LLM-based content-quality stage, gated so the most expensive stage only ran on inputs that had already cleared cheaper checks.
- Led the architecture and operation of a self-hosted LLM platform (Ollama, OpenWebUI) giving non-engineering teams safe, access-controlled use of AI tools for sensitive workloads.
- Set up and administered MongoDB as the database layer for a PayloadCMS-based content platform, including collection design and deployment configuration.
- Participated in code review and informally mentored junior engineers and students, sharing technical guidance and feedback.

Software Engineer, QED Solutions Ltd. (Nairobi)

April 2021 – July 2022

Backend engineer on Tendersure Africa, a SaaS B2B procurement platform used across multiple African countries.

- Built and maintained the platform with Django (backend) and Vue.js (frontend), integrating localised payment gateways across multiple African markets, handling signed callback verification, idempotent retries, and reconciliation.
- Managed and administered core infrastructure on AWS and DigitalOcean across EC2, RDS, S3, Route 53, and CloudWatch; led implementation of EC2 Auto Scaling groups to improve system availability.
- Led planning and provisioning of infrastructure using Terraform, significantly reducing the time to provision new environments.
- Led migration of legacy applications from VM-based workloads to Docker containers deployed on Amazon ECS, decoupling deployment from infrastructure.
- Configured AWS CloudWatch for monitoring and SNS for alerting on EC2 and RDS instances, resolving incidents in real time; later introduced a Prometheus and Alertmanager stack for application-level metrics.
- Led adoption of CI/CD pipelines using Bitbucket Pipelines to streamline build, test, and deployment processes.
- Implemented AWS Lambda functions to automate report generation from the core Django application, reducing load on primary application servers.
- Led migration of the organisation's sites from WordPress to static hosting on AWS S3 and Lightsail, reducing infrastructure costs.

Software Engineer, Nakala Analytics Ltd. (Nairobi)

August 2020 – March 2021

- Built data analytics applications in Python and Flask with REST APIs over PostgreSQL, integrating data from third-party APIs for downstream visualisation.
- Built ETL pipelines integrating PostgreSQL, dbt for transformation, and Stitch for source replication.
- Architected and deployed JupyterHub using Docker and Docker Swarm for the data analytics team.
- Built frontend interfaces in Vue.js and React for internal data analytics dashboards.

Software Engineer, JobsearchKE (Nairobi)

September 2019 – August 2020

- Built the company's core platform jobsearchke.com from scratch using Django, PostgreSQL, and JavaScript.
- Architected and deployed application infrastructure on DigitalOcean using Docker, Nginx, and DigitalOcean Spaces.
- Built an automated campaign and marketing platform integrating Mailchimp and SendGrid APIs.

Education

Jomo Kenyatta University of Agriculture and Technology, BSc Telecommunications Engineering, November 2018

Open Source Contributions

Django: Bug fixes and documentation contributions to the framework.

CKAN: Extension and plugin contributions to the leading open-source platform powering government and institutional open-data portals worldwide.